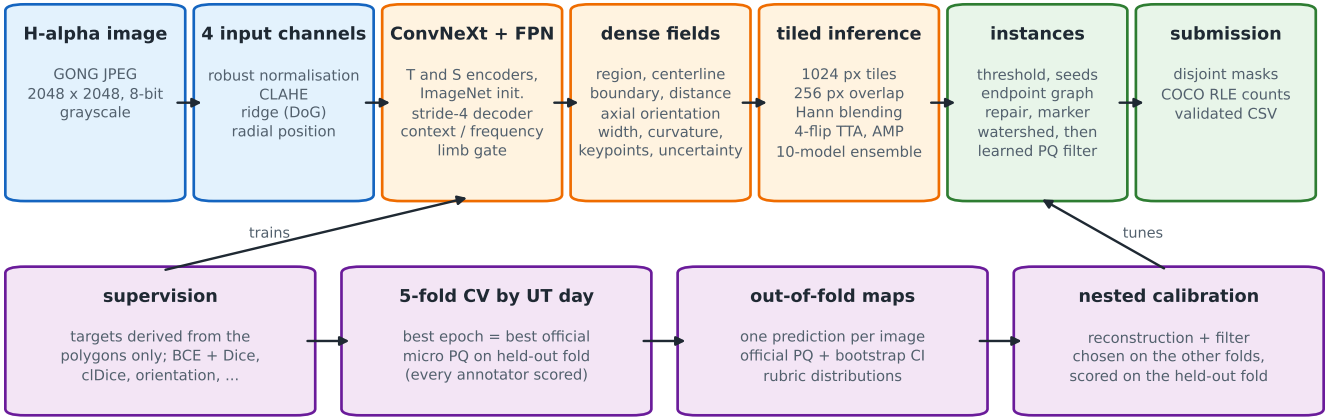
H-alpha image
GONG JPEG
2048 x 2048, 8-bit
grayscale
4 input channels
robust normalisation
CLAHE
ridge (DoG)
radial position
ConvNeXt + FPN
T and S encoders,
ImageNet init.
stride-4 decoder
context / frequency
limb gate
dense fields
region, centerline
boundary, distance
axial orientation
width, curvature,
keypoints, uncertainty
tiled inference
1024 px tiles
256 px overlap
Hann blending
4-flip TTA, AMP
10-model ensemble
instances
threshold, seeds
endpoint graph
repair, marker
watershed, then
learned PQ filter
submission
disjoint masks
COCO RLE counts
validated CSV
trains
tunes
supervision
targets derived from the
polygons only; BCE + Dice,
clDice, orientation, ...
5-fold CV by UT day
best epoch = best official
micro PQ on held-out fold
(every annotator scored)
out-of-fold maps
one prediction per image
official PQ + bootstrap CI
rubric distributions
nested calibration
reconstruction + filter
chosen on the other folds,
scored on the held-out fold